

Mishika Lunia

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Education

Indian Institute of Science Education and Research, Bhopal, BS in Electrical Engineering and Computer Science (GPA: 7.88/10.0)

May 2023

Experience

Research Assistant, KCDH, Ashoka University

June 2026 – Present

- Led development of a full-stack **multimodal data collection platform** using **React/Vite, FastAPI, and PostgreSQL** for offline-first clinical field research, supporting role-based access, opaque participant IDs, and site-scoped baseline/follow-up data.
- Implemented an **offline-first PWA** with IndexedDB drafts/outbox and idempotent synchronization for unreliable connectivity; built governed data-integrity workflows and hardened authentication using **AWS Cognito OTP, CSRF protection, Redis rate limiting, and activity auditing**.
- Deployed production infrastructure on **AWS (EC2, RDS, S3, Cognito)** using Docker Compose and Caddy HTTPS, with Alembic migrations, encrypted nightly backups, and multilingual support for **English, Hindi, and Nepali**.

Software Engineer, Orange Data Tech

Aug 2025 – June 2026

- Architected and deployed a production-grade **Adaptive Iterative RAG** system for legal compliance with multimodal document understanding.
- Built an automated remittance reconciliation platform using **Azure Service Bus, Blob Storage, etc.** and LLM pipelines, achieving **98% extraction accuracy** and eliminating manual invoice-payment matching.
- Developed full-stack B2B fintech features using **React, Django, REST APIs, and PostgreSQL**, plus a client-facing demo portal that contributed to new client onboarding.
- Integrated **Papermerge DMS** into an enterprise payroll platform, implementing **AES-256 encryption**, Redis-based rate limiting, and updated RBAC.

Research Assistant, DoMS, IIT Roorkee (Guided by: Dr. Gaurav Dixit)

May 2024 – Apr 2025

Research Focus: Impact of Government IT/ITES Policies on Corporate Profitability in Indian Markets

- Analyzed 105,000+ firm-year observations (2010–2023) to evaluate the causal impact of Indian IT/ITeS policies (**Digital India, IndiaStack/UPI, Make in India**) on firm profitability across Fintech, IT, Manufacturing, and Transport sectors.
- Implemented **TWFE-DiD** models with event study validation (± 4 -year windows) and **Arellano–Bond Difference GMM**; validated specifications using Hausman, Breusch–Pagan, Sargan, and AR(1)/AR(2) diagnostic tests.
- Applied **XGBoost**, Random Forest, and LSTM/GRU models for profitability prediction ($R^2 = 0.83$, **MDA = 80%**); income/expense ratio and firm size emerged as strongest predictors.
- Found no statistically significant causal effect of the policies on firm profitability once firm-level heterogeneity was controlled.

Certifications

Executive Postgraduate Program (Online), IIT Roorkee

Jun 2023 – May 2024

Data Science, AI/ML & Data Engineering,

Technologies

Languages: Python, R, Javascript, PostgreSQL

Technologies: PyTorch, Langchain, Hugging Face, Docker, REST API, FastAPI, Django, Azure, Azure AI Foundry, Azure Data Studio, Linux